Strathmore University
School of Computing and Engineering Sciences

Bachelor of Science in Informatics & Computer Science

ICS 3202: Artificial Intelligence

Project Deliverable 2 (CAT 2)

Start Date: 19th October 2026 **Due Date:** 6th November 2026

In your assigned groups for the semester project, you have identified a research problem to address that requires an application that uses a Machine Learning+Mobile App approach as the technical solution.

While developing the ML engine, a relevant algorithm needs to be trained, tested and validated on a sample dataset to meet the expected workflow in the final application. This deliverable focuses on synthesising the machine learning process for building the ML engine.

Instructions:

1. **As per your assigned groups for the semester,** identify the ML approach and algorithms that are relevant to the dataset & expected output of your application. Include this information in the shared spreadsheet of the semester groups.

2. In your notebook/development environment, critically detail the data cleaning process taken from exploration to the final dataset for modelling. As part of the EDA, derive at least 4 observations from the dataset.

3. Describe the modelling approach taken on the identified base model. Ensure you have applied appropriate measures as guided by other similar approaches to verify the validity of the results e.g. *applying elbow method for K-means clustering, cross validation for classification and regularisation for regression tasks*.

4. Determine other methods that can be used to further validate the results of the base model. A recommended approach would be to develop models that follow the same process and compare performances of each. See figure 6 of this research paper to get an idea of how this can be accomplished.

Submission Instructions

1. You will present your work to the instructor at scheduled times during our classes between **26th October and 6th November**. Your group will be scheduled on a specified date as indicated in the spreadsheet. **NOTE: attendance of all four classes is mandatory for all teams, regardless of the presentation date**.

    - For online classes, everyone in the class will join the regular session on Teams, and the presenters will generate and share separate meetings with the instructor.

2. **All members are expected to be present during presentation.** A score for teamwork will be assessed during the presentation. Ensure you present before the due date or your group risks losing marks for the entire CAT.

3. The key factor being assessed is your understanding of the data mining process from start to finish. The rubric that will be used is:

    - Teamwork (**10 marks**) - are all members present? did the group work as a team?

    - Presentation (**12 marks**) - does the presenter understand their section of the code? are they presenting it in a simple, digestible manner? Don't purpose to impress the instructor with machine learning jargon, but each group member should at least describe how they contributed to the work in a simple way.

    - Notebook (**8 marks**) - is the data mining workflow clear across the notebook? use of suitable commands? appropriate commenting? This will be assessed more in questions 2 and 3. Question 4 will be reviewed as the final and complete notebook at the end of the project. At this stage, a description of the validation approach is sufficient.

4. Once done, you will submit this in a Python Notebook titled *"Model_Base.ipynb*", where ***Model*** is the name of the ML model you were working on.

5. One group member will publish to GitHub in your repository named *SemProject_MLEngine* and the rest of the team members will fork the repository to

their individual accounts. (NOTE: for forking, the original repository needs to be set to public)

6. The notebook will be assessed as a marking point in the final stage of the project; therefore, it is essential for it to be in the repository before the due date of this CAT (6th November). ***The date of submission will be verified at the final stage of the project.***

Feel free to use any available resources for your work, however the rules against plagiarism apply. **Do not directly use ChatGPT or any other AI-generative text applications to develop your work**, this will be penalised strictly when detected.